
Rethinking Relational Evidence in Fixed-Candidate Retrieval:

Ablation Gains, Equal-Fact Alternatives, and Decision Advantage

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Abstract

A positive graph-feature ablation does not by itself establish that graph representation is irreplaceable or that a learned decision rule beats a simple graph-aware rule. We separate these claims for fixed-pool STaRK-Amazon retrieval. An 18-feature ranker on 384 held-out TRAIN queries (H) hits 278 answers, versus 264 for its fact-deprived text ablation and 263 for a candidate-shuffled graph control, but 279 for a top-linked-entity rule using the shared inputs. A distinct 20-feature adapted ranker on 768 other TRAIN queries (I) hits 568 versus 549 and 552 for its matched controls, but 573 for the rule. H and I evaluate different methods, not two confirmations of one method. Post hoc equal-fact text programs reproduce every rule decision and all six original graph columns; an H-informed structured-text learner ties the graph ranker at 278/384 on exposed H queries. These observations support a bounded relational-feature increment over text without relational facts, not graph-only informational value or learned-rule superiority. A new cohort is unscored; matched end-to-end resource comparison and human semantic review remain outstanding. This is a working draft, not an independently confirmed submission.

1 Introduction

Queries about a product and a named reference may require both attribute matching and relational evidence. A co-view or co-purchase edge can narrow a search, but does not establish that a candidate has the requested role or is compatible with the reference. In hybrid retrieval, an improvement relative to an ablation that lacks those edges can be mistaken for evidence that graphs are uniquely necessary, or that a trained graph ranker is the best decision rule. Those claims require different controls.

We ask three questions on the same fixed 20-candidate pools: (1) does matched relational evidence add to a ranker whose text branch has no relation facts? (2) can a text-side procedure with the *same facts* reproduce graph-driven decisions or features? (3) does learned fusion outperform a strong, simple rule with the same underlying candidates, rankings, reference links, and edges? The distinction between fact availability, representation and decision policy is the contribution of this empirical audit, not a new retrieval algorithm. It complements studies of semi-structured retrieval [Wu et al., 2024, Lee et al., 2025, Boer et al., 2026, Tao et al., 2026] and graph reranking [Dong et al., 2024, Li et al., 2025]; neither fixed-pool reranking nor matched ablations are claimed as firsts here.

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The two frozen Amazon evaluations yield a positive answer to the first question for their respective methods, but not to the third. Post hoc text replays challenge a claim of fact exclusivity without constituting a matched-budget LLM baseline. A separate exploratory MAG author task shows that an author-field text join can reproduce graph decisions on linkable queries; it is not a cross-domain confirmation. These distinctions matter most when a negative strong-baseline result would otherwise be hidden behind a successful ablation.

2 Related work and contribution boundary

Retrieval from text and relations. STaRK [Wu et al., 2024] supplies text-and-relation entity retrieval benchmarks; its synthetic and human query sources should not be interchanged. Our reported Amazon H/I queries come from the official TRAIN pool, and our fixed-pool Hit@1 cannot be compared numerically to the benchmark’s full-corpus TEST systems. HybGRAG [Lee et al., 2025] and Autofocus Retrieval [Boer et al., 2026] combine evidence discovery, constraints and reranking; GRASP [Tao et al., 2026] studies plan-guided graph retrieval and fusion. We freeze the initial 20 products and ask which *inference* follows from an additional set of relational features at the decision stage. Such a restriction removes important system behavior, notably candidate recall, index updates and open-ended reasoning. It is useful for attributing one decision change, not a replacement for those end-to-end systems.

Graph reranking and strong controls. G-RAG [Dong et al., 2024] already studies graph-based reranking of initial retrieval results. SubgraphRAG [Li et al., 2025] studies structural scores and matched-candidate or random-retrieval comparisons. Matching candidate pools, shuffling relations and applying LambdaRank [Ke et al., 2017] are therefore not method inventions. Here the additional diagnostic is an ordering of *claims*: first compare matched graph features to a fact-deprived text arm, then ask whether the same relation facts yield reproducible text-side actions, then compare learned selection to an explicit graph-aware rule. Only the first stage was an original H/I frozen ablation family; the equal-fact parser and structured-text learner were built post hoc. Treating the later controls as if they had been prospectively designed would erase the very research history that motivates this distinction.

Statistical evidence is not deployment evidence. The paired-query discordance test asks whether a fixed arm yields more benchmark fixes than harms under its null assumptions. It does not measure semantic compatibility, marginal dollars, or robustness to a new source of queries. The top-link rule consumes graph/link facts, so its higher hit count challenges the particular learned fusion decision, not the utility of graph access. Conversely, a text parser that can recreate support moments from recorded typed edges shows that those moments are not *informationally exclusive* to a graph data structure. It says nothing by itself about online text-model quality or compute parity. The contribution is therefore a scoped empirical audit of what the existing controls license one to say, not a proposed universal replacement for relational search.

3 Problem and evaluation roles

Given query q , frozen candidate pool C_q of size 20, and labeled answer set Y_q , a policy outputs $\pi(q) \in C_q$. Our outcome is

$$\text{Hit@1}(\pi) = \frac{1}{N} \sum_{q=1}^N \mathbf{1}\{\pi(q) \in Y_q\}. \quad (1)$$

The statistical unit is the query, not a candidate, graph world, LLM response, or pair of methods. Synthetic benchmark answer sets do not certify semantic negatives. We keep official TEST and human queries out of this study.

The graph ranker fits on 768 official TRAIN queries (A/B/F); 384 repeatedly used TRAIN queries (C/D/E) serve development. H comprises 384 different TRAIN queries sampled by a frozen sorted-ID permutation after method selection. The separate adapted 20-feature method additionally fits its semantic encoder on 1,024 disjoint TRAIN queries (S), then evaluates on 768 further TRAIN queries (I), with its comparison family fixed before query preparation. H and I are disjoint but differ in

method and supervision; both are now exposed. Neither is official TEST, an independent replication of the other, or an independent cross-dataset evaluation. Query disjointness does not imply disjoint products, graph edges, pretraining corpora, or freedom from historical research selection.

4 Matched inputs and competing decisions

BM25 and a frozen E5 retriever supply the RRF top-20 pool. Two LLM services rank those same candidates; one shared reference-extraction call identifies at most three named products per query. Each reference has at most three frozen linking options. For interpretation h and candidate v , let

$$U_q(v, h) = \frac{1}{m_q} \sum_{j=1}^{m_q} \mathbf{1}\{v \text{ is connected to reference } j \text{ in } h\}, \quad (2)$$

when $m_q > 0$; without a reference, all support statistics are zero. This proxy does not enforce direction, edge-type semantics, or product compatibility. An 18-dimensional LambdaRank fusion model [Ke et al., 2017] combines six symmetric two-rank statistics, six frozen semantic/text statistics (including a Qwen reranker), and six uniform or link-weighted world-support statistics. Its 12-dimensional text ablation retains ranks and semantic inputs but has no relation facts. A separate shuffled control permutes the six graph columns across candidates within each query, using one pre-specified seed for training and evaluation. These arms share the candidate pools, available rankings, supervision and ranker setup; the shuffle is not a randomization of the full knowledge graph and has not been repeated across seeds.

The strongest simple comparator first chooses the highest-weight link for each mention, then ranks candidates by its support, two-LLM RRF score and frozen candidate order in lexicographic sequence. With no references it reduces to RRF and the order tie-breaker. This *top-link* rule has access to the same underlying evidence, but not identical feature columns or the same minimum deployment cost as the learned model. In particular, the learned model does not have a hard support-first decision constraint. RRF and Borda require fewer components than the graph and Qwen paths when deployed alone. The adapted I method adds semantic adaptation and 20-dimensional controls; any difference between old and adapted methods also changes training data.

Feature construction and ambiguity. Let $\mathcal{M}_q$ be the nonempty set of strictly valid rankings, with $r_m(v) \in \{1, \dots, 20\}$. The six rank columns are normalized minimum, maximum, mean and range of ranks, mean reciprocal rank and fraction of first-place votes. These statistics are invariant to exchanging the two services; invariance of a feature builder is not invariance of service quality. The six nonrelational columns encode initial retrieval rank, query/title token overlap, query/product-text token overlap, clipped text length, a frozen Qwen yes-minus-no relevance logit, and its difference from the query’s maximum logit. No logit is treated as a calibrated probability.

For mention j , define a link option set R_{qj} with weights w_{qjt} . Title-exact ties receive uniform weights; otherwise weights are a softmax of frozen E5 similarities at temperature 0.05. For $h \in \mathcal{H}_q = \prod_j R_{qj}$, put $p_q(h) = \prod_j w_{qj, h_j}$. From $U_q(v, h)$ the six graph columns are uniform mean, minimum, maximum and standard deviation across worlds, followed by p_q -weighted mean and standard deviation. At most 27 worlds are explicitly enumerated; the product weights are a heuristic and not calibrated entity-linking posteriors. With no mention all six columns are zero. The support proxy treats either direction of also-buy and also-view as a link, discarding type-specific semantics. Thus an informative feature can still reward a wrong product role.

Each query contributes one group of 20 binary-labeled candidates to LightGBM LambdaRank: 100 trees, learning rate 0.05, at most seven leaves, minimum 20 examples per leaf, L_2 regularization one and ranking truncation at four. The frozen training configuration and label gain remain the authoritative specification. Scores choose the highest candidate within the full pool, breaking ties by frozen candidate order; the learned model does *not* restrict its decision to graph-supported items. For I, a BGE semantic encoder is adapted on S using up to four positive and additional negative training products per query, then frozen before the 768-query fusion fitting. Two extra semantic columns bring the adapted fusion to 20 features and its adapted text control to 14. Positive injection belongs only to encoder training, never evaluation retrieval. The adapted encoder adds supervision; old-versus-adapted scores do not isolate a single algorithmic change.

5 What can an ablation establish?

Coverage and selection are distinct. For a fixed query let $O_q(D) = \mathbf{1}\{D \cap Y_q \neq \emptyset\}$ and $H_q(\pi) = \mathbf{1}\{\pi(q) \in Y_q\}$. A ranker restricted to C_q satisfies $H_q(\pi) \leq O_q(C_q)$. If a graph-constrained policy selects from a support domain $E_q \subseteq C_q$, its error decomposes as

$$1 - H_q(\pi) = [1 - O_q(C_q)] + [O_q(C_q) - O_q(E_q)] + [O_q(E_q) - H_q(\pi)]. \quad (3)$$

The three nonnegative terms count missing candidate answers, answers excluded by the domain, and within-domain selection errors. This is an identity for the recorded labels, not a graph-correctness theorem. Our 18D fusion scores *all* 20 items and is not itself constrained to E_q ; the decomposition applies only to policies that select within their stated domain.

Equal facts do not imply equal computation. Let F_q denote the recorded candidate identities, ranks, link options, weights and edges. Both graph-feature extraction and a lossless typed textual serialization are deterministic functions of F_q . If a parser preserves those fields, composing it with the support computation reproduces the same support values and any deterministic rule using them. This is a constructive *representational* observation, not an assertion that a bounded-token language model will learn the parser, that its inputs fit a fixed context, or that text indexing and graph traversal have equal latency or cost. Fitting a separate lossy text learner is therefore a distinct empirical test.

Decision opportunity is label blind. For two fixed policies g, t on Q , put $D(g, t) = \{q \in Q : g(q) \neq t(q)\}$ and let F and B count queries where g alone and t alone hit, respectively. Directly from the definitions,

$$\text{Hit@1}(g) - \text{Hit@1}(t) = \frac{F - B}{|Q|}, \quad |\text{Hit@1}(g) - \text{Hit@1}(t)| \leq \frac{|D(g, t)|}{|Q|}. \quad (4)$$

Agreement in action forces agreement in hit status for *every* possible answer set; disagreement does not guarantee a different hit status. A label-blind minimum-disagreement gate can reject comparisons with little room for a meaningful difference before answers are opened. Passing the gate gives no evidence of superiority, calibration, or significance. Equations (3)–(4) are elementary bookkeeping tools; we claim no new theorem or population guarantee.

The estimand depends on the control. For two policies on the *same* recorded candidates, define the paired query contrast $\delta_q(g, t) = H_q(g) - H_q(t) \in \{-1, 0, 1\}$. Its observed average is $(F - B)/N$, rather than the difference between two independently sampled success rates. In the matched H and I ablations, t is a separately fitted ranker with graph columns withheld or shuffled. Thus a positive contrast establishes an improvement relative to that specified *training and feature intervention* on the observed queries. It does not identify a causal effect of access to a graph data structure: the relation facts are absent from the text-only control, while a same-fact text representation is a different intervention. Nor does the paired contrast identify the marginal value of adding graph access to a deployed retrieval pipeline, because the shared candidate set and upstream calls are held fixed by construction.

Why stronger conclusions do not follow. Consider any recorded fact set F_q and a lossless serialization $s(F_q)$. The composition $g \circ s^{-1}$ returns exactly the choices of a graph policy g wherever the serialization is parseable; this supplies a counterexample to a claim that the same facts *must* be represented as a graph to produce those choices. It does not supply an efficient learner: a bounded model might fail to parse $s(F_q)$, and generating the serialization may itself require graph traversal. Conversely, a graph feature model can gain over a fact-deprived text ranker yet lose to a support-first graph rule, because equal inputs impose no ordering on different policies’ decisions. The two frozen cohorts exhibit precisely that logical configuration. Resource parity, generalization and human product relevance each require their own measurement, not an inference from the paired hit counts.

6 Frozen results

What was frozen. The H method, 384-query sample, query-level denominator, ten-comparison family and response parsing policy were fixed before H query preparation. The development rule

Table 1: Hit@1 counts on disjoint official TRAIN cohorts. H evaluates the 18-feature fusion; I evaluates a distinct adapted 20-feature fusion. All methods share each cohort’s fixed 20-candidate pool. These are not repeated tests of a single method.

Policy	H / 384	I / 768
Two-LLM RRF	261	550
Max-world support	276	570
Top-link support	279	573
Text fusion without relation facts	264	549
Shuffled-graph fusion	263	552
Graph fusion (18D on H; adapted 20D on I)	278	568

Table 2: Selected paired results for the frozen primary method versus each control. The Holm families contain all 10 H and 15 I prespecified comparisons, not only the displayed rows.

Cohort	Comparator	Fix / harm	Net	Holm p
H	Matched fact-deprived text	15 / 1	+14	0.002594
H	Candidate-shuffled graph	17 / 2	+15	0.003279
H	Two-LLM RRF	26 / 9	+17	0.023952
H	Top-link support	13 / 14	−1	1
I	Matched adapted text	24 / 5	+19	0.003277
I	Adapted shuffled graph	21 / 5	+16	0.013717
I	Two-LLM RRF	37 / 19	+18	0.111207
I	Top-link support	20 / 25	−5	1

required positive gains over ten controls, at least five distinct fixes over RRF and no development sub-cohort RRF loss worse than two. That gate selected whether to run H, not a test statistic. The I method was separately frozen after S fitting and development, with 768 evaluation queries and a *different* 15-comparison family. Neither evaluation is an untouched official TEST set; previous experiments may have influenced the selected methods. Within each cohort all compared policies reuse the available rankings and fixed candidate pools. A strict ranking must contain every candidate exactly once. When one service fails parsing, the valid ranking is reused for all fusion methods; when both fail, the query remains in the denominator with zero fusion hits.

On H, the graph method exceeds matched text by 14 hits and shuffled graph by 15. One-sided exact paired discordance tests were adjusted by Holm over the *full ten* prespecified comparisons; adjusted p values are 0.002594 and 0.003279, respectively. The corresponding adjustment for RRF (+17 hits) is 0.023952; Borda makes the same choices as RRF here, not an independent replication. The graph ranker has 13 fixes and 14 harms against top-link (net −1, adjusted $p = 1$ for graph superiority). On I, the adapted graph exceeds its matched text by 19 hits (24 fixes, 5 harms; adjusted $p = 0.003277$) and the shuffled control by 16 (21 fixes, 5 harms; adjusted $p = 0.013717$), with Holm applied to *all 15* prespecified comparisons. Against top-link the adapted method instead has 20 fixes and 25 harms (net −5; adjusted $p = 1$). The adapted-vs-original graph difference on I is +5, not significant after the prespecified adjustment. These families do not correct for all method selection in the preceding research history; nonsignificance against the rule establishes neither equivalence nor noninferiority.

On H, 20,000 paired-query bootstrap resamples gave Bonferroni-adjusted approximate percentage-point intervals of [1.042, 6.771] relative to matched text and [−4.167, 3.646] relative to top-link. On I the analogous intervals are [0.5208, 4.6875] and [−3.2552, 1.8229]. Resampling assumes an appropriate query-sampling interpretation and is not a finite-population guarantee. Holm controls each fixed within-cohort family, not every adaptive choice made earlier in the research history.

For a fixed comparison of g against t , F is the number of g -only hits and B the number of t -only hits. Under an exchangeable null on the $F + B$ discordant queries, its one-sided exact tail is

$$p_{g>t} = \sum_{k=F}^{F+B} \binom{F+B}{k} 2^{-(F+B)}, \quad p_{g>t} = 1 \text{ if } F+B = 0. \quad (5)$$

Holm adjustment was applied to every raw test in each prespecified family, including comparisons not printed in Table 2. Paired bootstrap resamples share the same query indices across all arms

(20,000 draws per cohort); approximate Bonferroni simultaneous 95% intervals are reported where relevant. Neither computation removes dependence induced by the earlier iterative research process.

The frozen 20-item pools contain labeled answers for 354/384 H queries and 704/768 I queries, but this label oracle is not a deployed policy. H required 384 shared plans, 768 ranking requests and 7,680 local Qwen evaluations; three rankings failed strict parsing and remain in the denominator. I had three strict ranking parse failures among 1,536 ranking requests, also retained. These are query-level evaluations, not independent response-level samples or a matched end-to-end cost comparison.

7 Equal-fact alternatives and decision opportunities

After H and I became exposed, we serialized the cached reference options, their weights, typed and directed edges, both rankings, and candidate order as parseable text. A separate label-blind parser reproduced *all* top-link choices on H (384/384) and I (768/768). Independent scalar reconstruction of the six original relation columns matched 46,080 H and 92,160 I stored numbers within 10^{-10} . This establishes that these particular facts and the rule can be implemented from textual serialization; relabeling the same columns as “text” would not establish an alternative learner.

We therefore also tested an H-informed, post hoc structured-text learner: 18 lossy token statistics parsed from the same cached relation facts, combined with the original 12 semantic features, fitted with the original 768 query labels and LambdaRank setup. On *already exposed H*, it hits 278/384, the same as graph fusion (4 fixes, 4 harms; 358 identical choices), while top-link hits 279/384. This is a development result, not proof of equivalence, equal costs, an equal-token LLM text baseline, or independent validation. For exploratory MAG author retrieval, the author-field text join copied the graph path’s decisions on linkable TRAIN queries. MAG has neither an independent task evaluation here nor a transferable superiority result.

Identical action policies cannot differ in Hit@1 under any answer labels. Before scoring a new comparison, label-blind action disagreement is therefore a necessary check for decision opportunity, not a significance test. A frozen 384-ID prospective Amazon TRAIN queue exists, but no queries in that queue have generated scored actions: the launch preflight stopped before query-text access and external calls. We do not include it in any result table.

Where are the gains? An exploratory analysis defined graph activation after observing H/I as a maximum-support domain smaller than 20. Activation occurs for 55/384 H and 94/768 I queries. On H, the 18-feature graph fusion fixes 13 and harms zero relative to text within the 55 activated queries; the remaining queries account for a net gain of one. On I, the adapted method fixes 17 and harms one relative to its adapted text arm within the 94 activated queries, with a net gain of three outside. This accounting describes the already exposed samples; it is not a prespecified subgroup test or proof that a particular edge caused a successful prediction. On the same activated sets, top-link hits 37/55 H and 67/94 I, versus the respective fusion methods’ 32/55 and 59/94. An informative activated subset can coexist with an inferior learned decision policy.

Why is the rule hard to beat? Only 145 of the 768 queries used to fit the original fusion have a reference mention; 111 have nonconstant graph features, and 58 have both labeled positive and negative candidates within the maximum-support domain. On exposed I, the adapted fusion and top-link choose different products on 197 queries, but only 45 choices change Hit@1 status (20 favorable to fusion, 25 favorable to the rule). These counts are consistent with scarce decision-relevant supervision and harmful overrides, but do not isolate training coverage from representation, support-first policy, or distribution effects. They are not an additional evaluation set.

8 Limitations and research status

How the three questions interact. An ablation comparison holds supervision, candidate pool and much of the scoring stack constant while removing graph-derived columns. Its positive paired outcome identifies a benefit of that feature block *against that fact-deprived arm* on the frozen queries. It does not identify whether the information can be encoded as text, whether an LLM can use the encoded facts under a fixed token budget, or whether a different decision policy would have done better with the same facts. The rule replay shows constructively that textual serialization can repro-

duce a specific deterministic decision. The H-informed text learner goes beyond relabeling graph columns, but is post hoc and still not a prospective equal-resource study. Finally, top-link is a graph-using rule; its higher hit count is a failure of the *learned policy’s superiority claim*, not evidence that graph relations are useless. This separation is the empirical point of the paper: the strongest observation and the strongest counterexample address different questions.

Mechanistic interpretation is conditional. Under the observed link options and fixed pool, a support domain often retains a labeled answer. But a graph edge is merely an existence proxy: related accessories, alternate products and incompatible items can all be linked. The I action accounting makes the bottleneck concrete. The adapted learner and top-link disagree on 197/768 selected candidates, yet their benchmark hit states differ on only 45 queries. To beat the rule by selecting the learner selectively, one would need to recover enough of its 20 unique fixes without triggering its 25 unique harms; this is a sample accounting condition, not an identified way to build such a selector. Enlarging candidates, improving reference linking, training a role-aware score and imposing a support-first constraint would modify different components. None has been shown in this study to reverse the frozen strong-rule comparison under new evaluation.

These comparisons isolate an information increment against *fact-deprived* text, not against a trained text system with identical relation facts and matched indexing, latency, peak memory and price. Post hoc cached-feature and paired construction microbenchmarks exclude retrieval, model loading, reference API requests and Qwen; they cannot establish deployment efficiency. Top-link uses graph facts too, and its stronger outcomes do not mean that a text-only retriever dominates graph retrieval. The observations are about fixed-pool selection, not full-corpus end-to-end STaRK performance.

H/I are exposed TRAIN cohorts under distinct prespecified families, not independent final confirmation; synthetic qrels are not human semantic judgments. A prepared blind review of H/I decisions has no completed human labels. The new cohort remains unscored pending enforceable service spending limits, frozen matched-resource protocol, source and provenance checks, and the label-blind action gate. Future work must accept negative outcomes, assess robustness beyond a single shuffle seed, compare equal-fact methods on new queries with matched end-to-end resources, and complete human review. The current evidence archive is locally auditable but is not yet an approved, redistributable public reproduction package.

9 Conclusion

Relational features add value over text without relation facts in the two studied Amazon settings, but that increment does not establish a uniquely necessary graph representation or a learned decision advantage over the shared-input top-link rule. The three claims require distinct controls. Until new-cohort scoring, matched end-to-end comparison and semantic review are complete, this is a bounded empirical working draft rather than an independently confirmed or submission-ready superiority claim.

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A Evidence roles and reproducibility notes

The present manuscript is a condensed English review draft of the current Chinese working paper, not a second experimental result. Table 3 prevents query counts from being reused as independent confirmations. Earlier validation experiments tested different repair policies and are not pooled with H/I. All numerical claims in Tables 1 and 2 refer to archived frozen actions and query-level results. Independent readers need access to those records and authorized data before claiming a full reproduction; a local consistency check is not independent retraining.

Table 3: Data-use roles. Counts refer to queries, not LLM calls or candidate products. No official TEST or human-query answers are evaluated.

Set	Queries	Use and evidential status
A/B/F (TRAIN)	768	Original fusion fitting
C/D/E (TRAIN)	384	Repeated development; exposed
H (TRAIN)	384	Frozen 18D evaluation; now exposed
S (TRAIN)	1,024	Adapted semantic encoder training
I (TRAIN)	768	Distinct frozen 20D evaluation; now exposed
Prospective ID queue (TRAIN)	384	IDs frozen; no scored actions
Official TEST / human	1,642 / 81	Not used for reported results

The model comparison was fixed before H query preparation, with 384 plans and 768 rankings budgeted; the adapted comparison on I specified 768 plans and 1,536 rankings. Strict parsing failures stay in denominators, with available ranks reused consistently across same-cohort arms. The exact one-sided paired tests compare fixes to harms across query IDs, with Holm adjustment within each frozen family. A same-result action is never an additional successful replication. The structured-text arm, activation subgroups, and the MAG author-field join are explicitly post hoc or TRAIN developmental; none enter the original H/I comparison families. The staged prospective ID queue does not permit retroactive selection or scoring. Its zero-call preflight and lack of a metered cost stop prevent us from claiming that confirmation has started.

Publication requires separate review of style-year rules, author approvals, data/service-output redistribution rights, artifact privacy, checklist answers and the eventual scientific claim. Anonymous styling and local typesetting do not constitute a conference submission.

B Evidence and completion status

This appendix separates protocol details, equal-fact comparisons, local resource accounting, exploratory MAG observations and the complete frozen Amazon comparison families. Each has a different evidential role: a reconstructed action is not a new labeled evaluation, a local timing is not a deployed cost, and a frozen comparison family does not account for earlier choices in the research program.

C Claim-to-evidence trace and reproducibility boundary

The local analysis tree contains archived protocols, request and action snapshots, source hashes and validation reports. A reader can verify arithmetic without rerunning models, but cannot independently reproduce end-to-end behavior from the PDF alone. Table 4 specifies what evidence should be inspected for each claim. Paths are workspace relative; access to a path is not a license to redistribute its contents. This manuscript’s English wording must remain consistent with the more detailed Chinese working draft and the read-only evidence validator.

Table 4: Audit targets under `output/analysis/`. Numeric IDs identify the corresponding numbered report; H, I and queue IDs denote frozen experiment directories. A local validation is not external replication.

Claim	Evidence ID	What it checks
H 18D comparison	H 20260922	Protocol, actions, 384 hits, ten tests
I 20D comparison	I 20260923	Different method, 768 hits, 15 tests
Same-fact rule replay	Report 66	H/I label-blind action identity
Six graph columns	Report 67	Scalar tolerance and input hashes
H text learner	Report 68	Post hoc fit, gate, actions
Local resource scopes	Report 70	Separate timed boundaries
Unscored ID queue	Queue v0	Zero-access original protocol
Stopped launch	Launch v3	No query/call clearance
Human review design	Report 65	Prepared rubric, zero labels

H and I are the directories `multiview_transfer384_20260922/` and `disjoint_transfer768_20260923/`. Numbered IDs refer to the matching Markdown report prefix in the analysis directory. Queue v0 and Launch v3 refer to `amazon_candidate_cohort_20260925_v0/` and `amazon_prospective_launch_20260925_v3/`, respectively.

The read-only command `perl -w paper_evidence_check.pl`, from the repository root, checks H/I summary and paired rows against frozen result files, original source and input hashes for the H text pilot, candidate-ID disjointness and zero-access status, and the two distinct H resource benchmarks. It does not fetch the official dataset, rebuild the BM25/E5 indices, reproduce service outputs, retrain Qwen, run a blinded reviewer, or prove that outside access to the candidate suffix never occurred. Some raw records and model files are locally archived under an ignored analysis tree; the project has not delivered a cleared public package. The generated PDF and all third-party styles are reading artifacts, not a substitute for inspectable authorized sources and exact version pins.

What a third-party reproduction would have to specify. At minimum: the STaRK revision and splits, full-text and graph schema, hashes of every source/input/model, the lexical query tokenizer, BM25/E5 initial retrieval and tie ordering, exact prompt and model IDs, per-request caps and error semantics, entity-linking options/weights, Qwen scoring revision, fit IDs and hyperparameters, 20 candidate IDs for every query, action serialization, and a rule for missing responses. A new run must specify whether API usage is billed and which units are returned by the provider. A local syntax or checksum check does not validate the truth of the benchmark labels or the legitimacy of redistributing the underlying responses. Independent execution may produce different API responses if the pinned service implementation is unavailable; in that case the difference must be reported rather than silently substituted with another model.

D Human semantic review is a separate estimand

The official synthetic qrels measure agreement with the benchmark answer construction, not independent assessment of product role, compatibility or intended purchase. The already prepared H/I blind review draws per cohort 48 graph-versus-top-link *disagreement* queries and 24 queries on which they agree: 144 tasks in all. H has 73/384 differing actions and I has 197/768; the sample is chosen from these exposed TRAIN cohorts, so any eventual semantic result remains post hoc and cannot transform them into an untouched confirmation. Each task displays at most the two products

selected by the studied policies. It does not expose all 20 retrieved products and cannot estimate full-pool human recall.

Two real independent reviewers should receive separately shuffled blinded forms containing the same source query, chosen-product text, tags and recorded reference-edge facts. They should not be shown policy names, frozen benchmark labels, previously computed outcomes or the private mapping from displayed options to policy identities. Each must label product role match, explicit textual conflict, relational-condition support and overall suitability for *each* displayed option, using yes/no/uncertain with a cited justification. An also-buy/also-view edge is evidence of co-occurrence, not proof of semantic compatibility; missing graph evidence is not a verified negative. Separating the two reviewers' access from the workspace containing the private mapping is an operational requirement, not something that a validator script can ensure.

Before unblinding, validate completeness and the two identities, archive first-pass judgments, report uncertain rates and categorical agreement, and perform a separately recorded adjudication. Only then join the graph-versus-rule actions to suitability judgments, reporting fixes, harms, both-yes, both-no and unresolved separately for H and I. Treat unresolved paired differences conservatively with their possible $[-1, 1]$ contribution rather than dropping them. The agree-action tasks serve as rubric and baseline checks, not evidence of a policy difference. The disagreement-stratified sample is not IID from all benchmark queries. These are analysis instructions; as of this draft *no human annotations have been completed*. Even completed judgments cannot replace a new, independently frozen algorithmic evaluation.

E Prospective study: stop conditions, not outcomes

A later preflight froze 384 official TRAIN IDs selected from the sorted-ID permutation at positions 4672–5055. Its immutable v0 record sets paid calls to zero and forbids opening the new query text and answers. It proposes a graph policy, the separately H-designed structured-text learner and top-link on identical 20-candidate pools, with a label-blind minimum net action-disagreement threshold of five against *both* controls. This gate is necessary for a meaningful Hit@1 contrast but cannot predict the sign of the contrast. Passing it would not retrospectively remove H's influence on text-arm design.

User authorization later specified a task-bounded maximum of 1,152 external requests, conditional on a complete protocol and on the action gate before reading answers. The archived executor caps the attempt count and per-request output tokens and verifies observed service IDs, but records input usage and a provider cost field only *after* each call. It does not enforce a cumulative monetary hard stop before dispatch, and the cost field's billing unit and an explicit finite monetary ceiling remain unresolved. The separate launch preflight therefore returned STOP: no new-query text, answers or paid models were accessed by that attempt. Neither a user's allowance estimate nor a locally frozen ID list is proof that the cost-control or provenance conditions have been met.

To turn that queue into a real prospective experiment, one must first resolve remaining possible exposure outside known local metadata, verify the original service models without disclosing credentials, bind the action producer and its input/model files to an immutable protocol, implement enforceable token and monetary limits, and specify abstention and missing-call treatment. Only then may the approved query-only action generation run. The independently fitted equal-fact text arm should use the same cached edge/link facts, candidate pool and supervision and declare whether the target is pure quality at fixed facts or quality per measured resource. If full cold/warm lookup, comparable index construction and updates, token counts, latency, RAM/VRAM and amortized costs cannot be measured, that run cannot establish deployment superiority. Scoring any answers remains conditional on an archived label-blind action file that passes both predeclared gates. An unfavorable outcome must be reported without extending the queue or changing the policy.

This leaves four independent completion criteria for a mature paper: (i) an eligible prospectively scored comparison under correct model/budget guards, (ii) genuinely matched end-to-end resource measurements or an explicitly quality-only claim, (iii) actual independent human semantic judgments with reviewer separation, and (iv) author-approved reproduction rights and a redistributable evidence bundle. None of these is manufactured by typesetting.

F Implementation, data and frozen decisions

Dataset and candidate universe. The local STaRK-Amazon snapshot contains 957,192 products, 1,035,542 nodes, and 5,382,103 stored directed edges. Stored edge counts depend on the snapshot’s counting convention; they need not match a graph statistic reported for another revision. The full graph contains also-buy, also-view, brand, category and color edges, but the support proxy in this paper uses the first two types. The official synthetic query split contains 5,910 TRAIN, 1,548 validation and 1,642 TEST queries, plus 81 human queries. Synthetic query generation uses both text and graph information; that design may favor relational features relative to naturally sampled tasks. Product texts, graph structure, queries and model pretraining corpora can overlap even when evaluation *query IDs* are disjoint. A product not in the labeled answer set is a benchmark non-hit, not necessarily a human verified semantic negative.

Retrieval, ranking and failure handling. Frozen full-corpus BM25 and E5 ranks are joined using RRF with rank constant 60, and the first 20 candidates are retained. This candidate retrieval rank is distinct from the two subsequent LLM ranks of those 20 candidates. Both the learned fusion and the top-link rule use the same frozen candidate order as the ultimate tie-breaker. The two service IDs recorded for H are gpt-5.6-sol and gpt-6-astra; the identity check uses reported service usage rather than model self-report. The reference plan is a separate shared request to the latter service, limited to three literal product mentions in the query. An exact title match is attempted before dense E5 link proposals. Each LLM sees at most 512 o200k_base tokens of product text plus capped tags, but the E5 retrieval truncation and Qwen effective contexts are not all identical token windows. No claim of equal input length across encoders follows from sharing product identities.

A valid ranking contains every candidate exactly once. No ranking is silently repaired by appending absent products, and no parsed failure is removed from the query denominator. All arms in a cohort use the same usable service outputs. H had valid reference plans for all 384 queries and 765 valid ranking responses out of 768; I had valid plans for all 768 and 1,533 valid rankings out of 1,536. The original 18D fusion on H scores 7,680 product/query pairs locally with a frozen Qwen3-Reranker-0.6B yes-versus-no logit difference. Calls to obtain ranks and reference plans are shared by top-link, but the rule does not require Qwen prediction or learned-model inference to make its decision. A same-query prediction cost measured after all upstream features are cached cannot be used as its deployment cost.

Six rank and six nonrelational columns. For each valid model m , write $r_m(v)$ for its rank and $M = |\mathcal{M}_q|$. The rank block actually used on the fixed $K = 20$ pool is

$$\begin{aligned} \phi_R(q, v) = & \left[(\min_m r_m(v) - 1)/19, (\max_m r_m(v) - 1)/19, \right. \\ & (M^{-1} \sum_m r_m(v) - 1)/19, (\max_m r_m(v) - \min_m r_m(v))/19, \\ & \left. M^{-1} \sum_m 1/r_m(v), M^{-1} \sum_m \mathbf{1}\{r_m(v) = 1\} \right]. \end{aligned} \quad (6)$$

The nonrelational block holds normalized initial retrieval rank; query/title and query/text term overlaps; a capped normalized product-text length; the frozen Qwen relevance logit difference $z(q, v)$; and $z(q, v) - \max_{u \in C_q} z(q, u)$. The Qwen instruction distinguishes a requested purchase from a reference item already owned by a user, but its score is not a verified role or calibrated probability. The text ablation is $[\phi_R; \phi_T]$, the graph method appends ϕ_G , and shuffle preserves the six graph columns’ marginal values while moving the candidate association as one block within each query. Shuffling is controlled by a fixed seed and query identifier. Only one prespecified shuffle realization was evaluated; even a successful paired comparison does not estimate variance across many shuffles.

Six graph columns and policy. For options R_{qj} , the similarity softmax is

$$w_{qjt} = \frac{\exp(s_{qjt}/0.05)}{\sum_{u \in R_{qj}} \exp(s_{qju}/0.05)}, \quad p_q(h) = \prod_j w_{qj, h_j}. \quad (7)$$

Exact-title option ties instead receive uniform weights. If $A(v, t)$ denotes the observed existence of an also-buy or also-view edge in either direction, $U_q(v, h) = J_q^{-1} \sum_j A(v, h_j)$ for nonzero J_q . We

then take $\phi_G = [\mathbb{E}_{\text{unif}} U, \min U, \max U, \text{Std}_{\text{unif}} U, \mathbb{E}_{p_q} U, \text{Std}_{p_q} U]$. These moments flatten typed directed records into a coarse undirected existence indicator for scoring. The fact serialization used by the equal-fact controls retains the underlying edge types, directions and duplicates instead. This distinction matters: the six coarse values are not a semantics oracle. If no mention is identified, the whole graph block is zero.

For top-link let $h_j^* = \arg \max_{t \in R_{qj}} w_{qjt}$, resolving a link tie by frozen option order. Its lexicographic action is

$$\pi_{\text{top}}(q) = \arg \max_{v \in C_q}^{\text{lex}} \left(U_q(v, h^*), \sum_{m \in M_q} \frac{1}{60 + r_m(v)}, -\text{rank}_{C_q}(v) \right). \quad (8)$$

The second and third entries matter only after support ties. This is not a “weak text-only” baseline: it shares links and graph data with the graph fusion, but uses a different decision constraint. Max-world support replaces the first entry by $\max_h U_q(v, h)$, with the same two-rank tie-breaker. Its 276 H and 570 I hits are below top-link but above either learned graph method on I. A model trained without the top-link support column cannot be declared to have learned the support-first policy merely because the underlying facts were available.

F.1 Equal-fact controls: what is and is not tested

Rule and feature replays. The post hoc rule control serializes each query’s frozen candidate order, rankings, reference option IDs, option weights, and all recorded connected candidate/option edge IDs, types, directions and duplicates as UTF-8 structured text. An independent parser then selects each top link, recomputes its support and applies the same lexicographic RRF tie-break. It agrees on 384/384 H actions and 768/768 I actions; only 68 H and 138 I queries contain a named reference. The serialized per-query size spans 215–8,480 H bytes and 228–3,891 I bytes. These counts specify *bytes*, not a token budget for a model. A second label-blind replay enumerates all linked-option worlds and reconstructs the original graph block to absolute error at most 10^{-10} for every one of the 46,080 H and 92,160 I scalars (queries times 20 candidates times six columns). That construction can be performed in either a graph-oriented or text-parsed program. It does not establish that an off-the-shelf language model, given the same tokens, will learn an equivalent policy.

An H-informed learned text arm. To avoid calling a mere reserialization of six graph columns a learned text baseline, a separate H development experiment feeds the losslessly serialized *facts* to a parser that constructs 18 lossy lexical statistics per candidate. These are combined with the original 12 nonrelational features and fitted with the same 768 fitting queries, query groups, labels, LambdaRank settings and tree count. Original text and original graph models were retrained in the same CPU session and replayed their frozen H actions. The new text model makes 278/384 H hits, against 278/384 graph and 279/384 top-link. Relative to graph, its actions agree on 358 queries; of 26 disagreements only eight change benchmark hit status, four in each direction. Relative to the rule there are 13 fixes and 14 harms. The fit IDs and H query IDs are disjoint, but the design of this text arm was chosen after H outcomes had already been inspected. A tie in this exposed-cohort outcome is not evidence of equivalence or a matched-cost performance guarantee. The parser is lossy despite lossless input serialization, and it is neither an LLM-baseline evaluation nor an adapted-I semantic-encoder control.

Why input equality has several meanings. We distinguish the underlying fact set, its textual or graph-oriented representation, model supervision, candidate pool, upstream ranking calls, and measured deployment resources. The original text12 ablation shares a candidate pool, rankings, semantic scores and label budget, but does *not* have the link and edge facts. The post hoc parser has those facts, but is a deterministic program rather than a matched token/latency LLM. The H structured-text learner has the facts and original fit labels, but was designed after exposure and uses lossy lexicalization; there is no corresponding I adapted-text same-fact learner. The top-link rule has underlying facts but differs in features and support-first action logic. Conflating these controls would turn one valid ablation result into an invalid graph-exclusivity claim.

F.2 Measured local resources versus deployment cost

Two separate exploratory benchmarks reused *exposed H*; neither read new query answers or assessed quality on a new cohort. The first used seven serial repeated predictions over 384 queries and 7,680 candidates, alternating graph and text arms with one prediction thread. Its graph input features had already been cached. The text arm separately serialized relation facts into JSON and parsed its 18 columns before model inference. Both models reproduced their original H actions on each repeat. Table 5 separates these stages; timings are descriptive local measurements, not confidence intervals or end-to-end deployment comparisons.

Table 5: Exposed-H local resource observations, seven runs per stage. Different protocols must not be compared across the panel boundary.

Stage and measurement boundary	Median (s)	Range (s)
Text JSON serialization and parse; no index	0.550701	0.547480–0.756351
Graph model prediction; graph features cached	0.023100	0.022765–0.030991
Text model prediction; text features parsed	0.023335	0.023135–0.031144
Graph six-column construction, <i>preloaded</i> facts	0.126542	0.125997–0.128655
Text JSON and 18-column construction, <i>same preloaded</i> facts	0.113578	0.112287–0.114158

The second benchmark independently and symmetrically constructs graph and structured-text columns from the *same preloaded* H case and link-option data, seven serial repeats with alternating arm order. Its graph producer exactly recovers the 46,080 frozen graph scalars on every repeat. The first benchmark includes Python `tracemalloc` while timing text serialization; it also records a 4,311,096-byte median Python allocation peak during that parse, *not* process RSS. It is invalid to compare its 0.550701-second text parse directly with the other run’s 0.126542-second graph construction as an efficiency ratio. The construction outputs also have different feature dimensions.

Both benchmarks omit full-corpus index build/update, BM25/E5 retrieval, LLM reference generation and two ranking requests, entity-linking execution, Qwen scoring, model loading, process/VRAM peak and controlled cold/warm end-to-end latency. The recorded service cost fields have unverified units; dollars cannot be reconstructed reliably from them. Machine resources were not exclusively reserved to certify same-time-budget deployment. Neither arm is shown to be cheaper or noninferior under an actual matched end-to-end resource budget.

F.3 Exploratory cross-task and negative-control evidence

The exploratory STaRK-MAG author task retrieves papers, not Amazon products, and neither Amazon model is transferred to it. On a selected 1,024-query MAG TRAIN development sample, author-writes reranking hits 376, author-enhanced BM25 363 and a mismatched-support control 299. Yet exact matching against the same index’s author text field reproduces every graph-rule action on its 122 linkable queries. This supports a counterexample to a *representation-exclusive* interpretation of that particular graph path; it is not independent cross-dataset confirmation of Amazon’s learned fusion.

Other MAG TRAIN pilots supply deliberately negative diagnostic evidence. On 259 disjoint TRAIN queries, an external-author-ID graph profile hits 168, exact-name text 185 and BM25 153; these graph and text profiles are *not* strict equal-fact representations. On 434 collaborator queries a graph path and its named-text comparator differ on only four top-1 actions. On 512 institution queries, graph and same-fact text prompts differ on only nine actions. Those latter two comparisons failed their prespecified label-blind disagreement gates; their answers were not opened and no hit rates for them are claimed. The prompts do not have exactly equal lengths even when they share facts and model access. By Eq. (4), any Hit@1 difference on their fixed cohorts is at most 4/434 or 9/512 respectively, regardless of which policy would benefit. We did not evaluate MAG official VAL/TEST or human queries for these paths.

G Full frozen comparison families

Tables 6 and 7 show every prespecified comparator, not only the favorable rows in the main paper. The first column identifies a different policy on the *same* cohort: one model’s raw ranking, a single-model LambdaRank (LTR), a linked-gain single-model postprocessor, rank aggregation, or a graph-support rule. These policies can differ in minimal deployment cost even when evaluated on the same cached candidates and service outputs. In particular, rank-only policies need not run the extra Qwen scorer or reference-extraction path if deployed independently. Each table uses a *single* numerator (the frozen graph method) and a query denominator fixed before result inspection. A negative net count is a result, not a missing experiment.

Table 6: Complete H family ($N = 384$; primary: 18D graph fusion, 278 hits). Fix/harm is primary-only hit/comparator-only hit. Adjusted p uses all ten one-sided paired comparisons; rounding is for display only. LTR and linked postprocessors use the indicated single ranking.

Comparator	Hits	Fix/harm	Net	Holm p
Text fusion (12D)	264	15/1	+14	0.002594
Shuffled fusion (18D)	263	17/2	+15	0.003279
RRF	261	26/9	+17	0.023952
Borda	261	26/9	+17	0.023952
Single-model LTR 0	270	18/10	+8	0.554800
Single-model LTR 1	273	15/10	+5	1
Single-model linked 0	273	17/12	+5	1
Single-model linked 1	277	14/13	+1	1
Max-world support	276	14/12	+2	1
Top-link support	279	13/14	−1	1

The H training pool consists of A/B/F, not H labels; development on C/D/E was used to choose the method. The 384 H queries were held out *at the time of that freeze*, not across the complete research history. RRF and Borda coincide in this cohort, making their two rows two controls in the fixed family but not independent empirical replications. The apparent positive difference over maximum-world support is only two queries. The top-link control beats the main graph method by one hit; its nonsignificant comparison must not be reversed into an equivalence claim. Complete query-level action and hit records reside under the frozen H results archive (see Appendix C).

Table 7: Complete I family ($N = 768$; primary: adapted 20D graph fusion, 568 hits). The older 18D text/shuffle rows do *not* share the adapted encoder’s additional supervision, unlike adapted text/shuffle.

Comparator	Hits	Fix/harm	Net	Holm p
Old 18D graph fusion	563	17/12	+5	1
Old text fusion	542	33/7	+26	0.000296
Old shuffled fusion	543	33/8	+25	0.000729
RRF	550	37/19	+18	0.111207
Borda	550	37/19	+18	0.111207
Single-model LTR 0	553	43/28	+15	0.383695
Single-model LTR 1	562	34/28	+6	1
Single-model linked 0	555	42/29	+13	0.538695
Single-model linked 1	565	33/30	+3	1
Max-world support	570	24/26	−2	1
Top-link support	573	20/25	−5	1
Matched adapted text (14D)	549	24/5	+19	0.003277
Adapted shuffled graph (20D)	552	21/5	+16	0.013717
Original adapted-encoder score	561	17/10	+7	0.743366
Raw adapted model ranking	431	167/30	+137	$< 10^{-20}$

Five I comparisons are positive after the specified adjustment: matched adapted text, adapted shuffle, old text, old shuffle, and the standalone semantic score. The last three are not substitutes for an equal-supervision graph comparison. The raw adapted model ranking is included, much weaker control, not evidence of strong-baseline superiority. Neither RRF nor either graph-support rule is among the

five positive adjusted comparisons. The adapted model is not significantly better than the old 18D graph model on I, so the additional encoder-training step itself has no isolated confirmed gain. This family cannot be pooled with H’s ten tests or interpreted as 1,152 repeated trials of one policy.

H Derivations and finite-sample scope

Error accounting. For a policy required to select from $E_q \subseteq C_q$, the three terms in Eq. (3) are nonnegative because $H_q(\pi) \leq O_q(E_q) \leq O_q(C_q) \leq 1$. Summing the three brackets telescopes to $1 - H_q(\pi)$. Neither $O_q(E_q)$ nor $O_q(C_q)$ can be computed without the benchmark labels, and neither is an attainable deployed policy. The full-pool oracle upper bound on hit counts is 354/384 for H and 704/768 for I. Thus the 18D graph fusion’s 106 H misses include 30 without any labeled answer in the fixed pool and 76 with an answer present but not selected. The adapted fusion’s 200 I misses likewise split into 64 unavailable and 136 within-pool selection failures. This does not identify which failures arise specifically from the graph.

Paired outcomes. For each query define $d_q = H_q(g) - H_q(t) \in \{-1, 0, 1\}$. Its nonzero terms are the F fixes and B harms, hence $\sum_q d_q = F - B$. If two policies return the same candidate, both hit indicators coincide for every possible Y_q , so $d_q = 0$ whenever $q \notin D(g, t)$. Thus $|\sum_q d_q| \leq \sum_{q \in D} |d_q| \leq |D|$, proving Eq. (4). This bound is about the *observed fixed queries* and policies, not a rate guarantee for future queries. The paired exact test in Eq. (5) conditions on the observed number of discordant outcomes. It needs the interpretation that under the null their two signs are exchangeable; Holm’s family control applies to those particular frozen tests, not to previous model search. A confidence interval crossing zero does not certify noninferiority. To test that claim one would need a justified margin and a separate prespecified procedure.

Matched-size, matched-rank-set expectation. Partition each fixed candidate pool into five consecutive bins of four by the two-LLM RRF order. In bin b , let $n_b = 4$ candidates include p_b benchmark positives, and let a support domain select k_b items. A control set uniformly selects k_b without replacement in each bin. Exactly $\binom{n_b - p_b}{k_b}$ of the $\binom{n_b}{k_b}$ possible choices avoid positives. Since bin draws are independent by construction,

$$\mathbb{E}[O_q(E_{\text{control}})] = 1 - \prod_{b=1}^5 \frac{\binom{n_b - p_b}{k_b}}{\binom{n_b}{k_b}}. \quad (9)$$

The usual convention assigns zero to a bin’s avoidance probability when $k_b > n_b - p_b$. Equation (9) is an exact expectation for this artificial matched-set distribution, not a model of how graph edges are generated. A more finely matched degree or within-bin rank control might change the descriptive comparison. On the *post hoc* activated subsets ($|E_q| < 20$), recorded graph-domain oracles total 43/55 H and 79/94 I; the corresponding matched-set oracle expectations sum to 25.50 and 44.52. Fractional values are expectations, not fractional queries or additional independently sampled observations. On those same subsets top-link hits 37 and 67, while the matched-set first-position expectations are 16.92 and 29.67. All these numbers were computed after H/I exposure; no new confirmatory p value follows.

A restricted improvement ceiling. Consider a hypothetical new policy that differs from top-link only on the activated set $\mathcal{A}$ and still selects from E_q there. For each activated q , $H_q(\pi) \leq O_q(E_q)$, while outside $\mathcal{A}$ the two policies have equal hits. Consequently its *observed* improvement is bounded by

$$\sum_q [H_q(\pi) - H_q(\pi_{\text{top}})] \leq \sum_{q \in \mathcal{A}} [O_q(E_q) - H_q(\pi_{\text{top}})]. \quad (10)$$

The right-hand side is six H queries and twelve I queries, each 1.5625 percentage points of its respective full cohort. These are label-oracle ceilings conditional on the specified domain and activation restriction; they do not bound policies that change choices outside $\mathcal{A}$ or select outside E_q , and they do not promise an attainable gain. For example, the studied full-pool fusion is *not* constrained by the restriction. No novel generalization theorem is asserted here.